Advanced Empirical Research Methods

MT Examined Problem Set - MT2025

Part 1

Exam Rules

1. Use the Julia programming language, and write your code in either Jupyter Notebook or VS Code.
2. Provide a cleanly formatted PDF that contains all answers and all results, as well as a separate set of files that contains all code to replicate your results. Export all figures and estimates to a separate folder called "results".
    1. All results must be replicable using your code.
    2. Adding plots to the pdf that you are not able to replicate using your own solution (say because you know the consumption policy is concave in wealth, but your algorithm gives you something convex) is considered cheating and leads to failure.
3. Do this exam on your own. Do not copy other student's code. If this happens, both students will fail the exam. So please do not ask a fellow student for this kind of favor.
    1. You can use code from your homework, build on chat GPT, etc., and use built-in functions, but your code must demonstrate that you understand what is happening.
4. If you find that the exam is missing assumptions to solve a problem (say I forgot to specify the discount factor), make assumptions that will allow you to proceed, and state clearly what these assumptions are.

Setup: Risk and Volatility

Time is discrete. Consider an economy with a unit continuum of heterogeneous households subject to labor income shocks. Households can save in a risk-free bond b that pays gross return $1+r$, and there is an ad-hoc borrowing constraint $b \geq -0.5$.

Households live forever and solve the following dynamic problem:

$$V = \max_{\{c_t\}_{t\geq 0}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\gamma}}{1-\gamma},$$

subject to a standard budget constraint:

$$\begin{aligned} b_{t+1} &= (1+r)b_t + y_t - c_t, \\ b_{t+1} &\geq -0.5. \end{aligned}$$

where y is household labor income, which is stochastic and takes on two values, high and low: $y_L = 1$ and $y_H = 2$.

Use the following parameter values in Table 1 to complete the exercises, where tol_HJB and tol_KFE refer to the tolerance chosen in the value function iteration and when solving for stationary distributions, respectively.

Table 1: Parameters

Parameter	Value
β	.85
γ	1.5
r	0.10
tol_HJB	10^{-6}
tol_KFE	10^{-6}

Exercise 1: Solve Standard Consumption-Saving Problem for different values of ρ

1. Suppose income follows a two-state Markov chain with symmetric persistence, $\rho_L = \rho_H = \rho$. Derive the stationary income distribution for values of $\rho = [0.0, 0.05, .2, .5, .7, .95, 1.0]$ where you start out by assuming an initial distribution $\mathbb{P}(y = y_L) = .2$. Plot the share of low-income households over a hundred year horizon for the different values of ρ in one plot with appropriate labels. Explain what you see: do the shares evolve differently over time? If so, why? Do they converge to the same stationary distribution or not? Explain in max 2 sentences.

2. As you vary ρ, explain for which value labor income is most volatile as measured by the variance of the change in log labor income in the cross-section of households. Does this particular value for ρ also generate the most risk for households? Explain in max 2 sentences.

3. Suppose you want to simulate this income process for 1000 households where 20% of households start out in the low income state and the persistence equals $\rho_L = \rho_H = .95$.
   Please set the seed to 1776. Simulate this income process for 8 years where we start counting from 1 to 8, i.e., you have one initial condition, and then 7 more draws, which gives you 8 years of data. To make the results perfectly comparable, use the following command to create the random matrix on which you base the simulation:

```
rand_mat = rand(1000, 8)
```

Plot the mean share of high types in your simulated data, and compare it to the evolution of the share of high types from the previous exercise with a continuum of households. Do you expect these two outcomes to agree exactly? Explain in 1 sentence.

4. With the simulated data in hand, compute the following regression coefficients based on model 1, which uses a simple OLS regression of current on lagged labor income, and model 2, which adds individual fixed effects into the model but is otherwise identical. Run these regressions in julia only using linear algebra and demeaning of variables (do not load additional packages to input the statistical model directly – I want you to do the work).

$$
\begin{aligned}
m1: &\quad y_{i,t} = \alpha + \beta_{\text{OLS}} y_{i,t-1} + u_{i,t} \\
m2: &\quad y_{i,t} = \alpha_i + \beta_{\text{FE}} y_{i,t-1} + u_{i,t}
\end{aligned}
$$

Based on the underlying structural model that governs the income process, which regression gets closer to the truth? Is there a systematic bias in the regressions relative to ρ? Explain in max 3 sentences.

Exercise 2. Risk and Aggregate Savings – A Decomposition

1. Holding the interest rate fixed at 10%, compute household asset supply implied by different values of $\rho = [0.0, 0.05, .2, .5, .7, .95, 1.0]$. Argue why computation fails for $\rho = 1$. Can you still figure out the stationary asset supply in this case?

2. Note that the exogenous income process together with the endogenous household consumption-saving choice induces a stationary probability distribution over income and assets denoted by $\mu(b, y)$ such that
$$\sum_{y \in \mathcal{Y}, b \in \mathcal{B}} \mu(b, y) = 1,$$
where $\mathcal{Y}, \mathcal{B}$ contain all possible realizations of income and asset states.

We will explore how the distribution $\mu(b, y)$ depends on the indirect impact of ρ through the solution of the policy function $c(b, y)$ and directly through the immediate impact of ρ on the income distribution. We will use this approach to decompose the effect of changes in ρ on aggregate household asset supply.

I use the case with $\rho = .5$ as benchmark, and denote the solution using the superscript .5 i.e., the solution from the household side consists of policy function, distribution, and aggregate asset supply $\{c_{.5}(b, y), \mu_{.5}(b, y), B_{.5}\}$

Denote the solution for $\rho = 0$ as $\{c_0(b, y), \mu_0(b, y), B_0\}$ and for $\rho = .95$ as $\{c_{.95}(b, y), \mu_{.95}(b, y), B_{.95}\}$.

a. Compute the counterfactual aggregate asset supply where you keep the consumption policy fixed at the benchmark but you allow for the income process to change. That is, what is the aggregate asset supply when applying $c_{.5}(b, y)$ in combination with the income process for both $\rho = 0$ and $\rho = .95$. This gives you two counterfactual aggregate asset supplies denoted as $B(\rho = 0, c = c_{.5})$ and $B(\rho = .95, c = c_{.5})$, respectively.

b. With these objects in hand, we pursue the following decomposition:
$$\underbrace{B(\rho = 0, c = c_0) - B(\rho = .5, c = c_{.5})}_{T_{\text{total}}} = \underbrace{B(\rho = 0, c = c_0) - B(\rho = 0, c = c_{.5})}_{T_1} + \underbrace{B(\rho = 0, c = c_{.5}) - B(\rho = .5, c = c_{.5})}_{T_2}$$
where in particular we want to decompose the total change T_{total} into term T_1 and term T_2. Construct a clean table with two rows and three columns. Row one considers the case of $\rho = 0$ relative to the benchmark, and column one shows T_{total} while column two and three report the percent accounted for by T_1 and T_2, i.e., $\frac{T_1}{T_{\text{total}}} \times 100$ rounded to one digit after the decimal point. Row two performs a similar exercise but for the case of $\rho = .95$.

c. Explain this seemingly puzzling result to me: How come that changing ρ alone seems to have an effect on aggregate savings, even though the individual consumption policy function is held fixed and the stationary share of high and low types is unrelated to the value of ρ? Answer in max 2 sentences.

Part 2

Exam rules

Use `R` and produce your answers by generating a PDF from an `R Markdown` file which includes your code, its output, and your answers to questions in a single document.

Please, be concise in your answers: each time you are asked to comment, it should be no more than two short paragraphs, unless specified otherwise. Wordiness that betrays a lack of understanding will be penalized.

Do this exam on your own. Do not copy other student's code. If this happens, both students will fail the exam. So please do not ask a fellow student for this kind of favor.

You can use code from your homework, build on ChatGPT, etc., and use built-in functions, but your code must demonstrate that you understand what is happening. Please refer to the guidance on AI on the Canvas page.

Finally, if you find that the exam is missing assumptions to solve a problem, make assumptions that will allow you to proceed, and state clearly what these assumptions are.

Exam Question

This question is based on Huber & Mellace (2015) and Mourifie & Wan (2017). You will need to consult both papers.

(Q1) Write an `R` function that uses `rmvnorm()` from the `mvtnorm` package to simulate `n` iid draws from the model given below, with arguments `n`, `alpha` and `beta`. Your function should return a data frame with named columns `D`, `Z`, and `Y`.

$$
\begin{aligned}
Y &= D + \beta Z + U \\
D &= 1\{\alpha Z + \epsilon > 0\} \\
\begin{bmatrix} U \\ \epsilon \end{bmatrix} &\sim \text{Normal}(0, \Sigma), \quad \Sigma = \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix} \\
Z &\sim \text{Bernoulli}(0.5), \text{ indep. of } (U, \epsilon)
\end{aligned}
$$

Define another function that computes the IV estimate based on Z, D, and Y. Use it to test the function you created above.

(Q2) Answer the following questions about the model from the preceding part.

a) Is the treatment D endogenous? How can you tell?

b) What is the distribution of treatment effects? What is the LATE in this model?

c) What is the role of β?

d) What is the role of α?

e) Which of the LATE assumptions does the model satisfy?

(Q3) Explain which LATE assumption is different between the tests proposed by Huber and Mellace (2015, hereafter HM2015) and by Mourifie and Wan (2017, MW2017). Explain the relevance of both tests for the literature on Treatment Effects.

(Q4) Write a function called `get_theta_HM()` to compute the sample analogues of $\theta_1, \theta_2, \theta_3, \theta_4$ defined in Equation (7) of HM2015. Your function should take a single input argument: a data frame (or tibble) with

columns named `D`, `Z`, and `Y` corresponding to the model from above. It should return a vector with four named elements: `theta1`, `theta2`, `theta3`, and `theta4`.

Recall equation (7) of HM2015 - the **four testable inequality moment constraints**:

$$H_0 : \begin{pmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \\ \theta_4 \end{pmatrix} \equiv \begin{pmatrix} E(Y|D=1, Z=1, Y \leq y_q) - E(Y|D=1, Z=0) \\ E(Y|D=1, Z=0) - E(Y|D=1, Z=1, Y \geq y_{1-q}) \\ E(Y|D=0, Z=0, Y \leq y_r) - E(Y|D=0, Z=1) \\ E(Y|D=0, Z=1) - E(Y|D=0, Z=0, Y \geq y_{1-r}) \end{pmatrix} \leq \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

(Q5) Check your function from the preceding part by generating 100,000 observations from the model in part 1 with parameter values $\alpha = 0.6$ and $\beta = 1$. You should detect a violation of the LATE assumptions. Calculate the Wald estimand. Does it equal the LATE? Repeat for $\beta = 0$. How do your results change?

(Q6) Repeat the preceding part for a variety of values of β until you find one for which the LATE assumptions are violated but you cannot detect a violation of the inequalities from the paper. Why is this possible?

(Q7) Write a function called `get_theta_MW()` to compute the sample analogues of $\theta(y, 0), \theta(y, 1)$ defined in Equation (5) of MW2017. Your function should take as input argument: a data frame (or tibble) with columns named `D`, `Z`, and `Y` corresponding to the model from above, and a scalar y. It should return a vector with two named elements: `thetay0`, `thetay1`.

Recall equation (5) of MW2017 - the **testable inequality moment constraints**, for any $y \in \mathcal{Y}$:

$$\begin{cases} \theta(y, 1) \equiv \mathbb{E}[p_1 D(1-Z) - p_0 DZ | Y = y] \leq 0, \\ \theta(y, 0) \equiv \mathbb{E}[p_0 1(1-D)Z - p_1(1-D)(1-Z) | Y = y] \leq 0. \end{cases}$$

where $p_k = \Pr(Z = k)$.

(Q8) Check your function from the preceding part by generating 100,000 observations from the model in part 1 with parameter values $\alpha = 0.6$ and $\beta = 1$. For example, you can plot `thetay0`, `thetay1` against the values of y. Can you find values of β where you find a violation from the MW2017 and not from the HM2015? Is the converse possible? Explain your answer.

(Q9) Load the data `STEM_data_for_adv_ERM` (here) from Problem Set 3 on returns to STEM education (see the description here).

Define:

- `D` as `stemdeg`: equals 1 if the respondent has a STEM degree.
- `Y` as `income_5yrs`: the hourly wage after five years of graduation (please trim the distribution from extreme values).

You wish to test two possible instruments for men and women separately:

- `Z1` defined as `binmothered`: equals 1 if the mother has tertiary education.
- `Z2` defined as `bin_prop_faculty_stem`: equals 1 if proportion of female faculty in the year and region of Abitur is above the national average.

Apply your functions `get_theta_HM()` and `get_theta_MW()` to the uploaded data. Do you detect any violations of the LATE model?

(Q10) We did not consider statistical significance when looking for a violation of the LATE model in the preceding part. Use the function `boot()` from the R package `boot`, along with your function `get_theta_HM()` from above to implement the "simple bootstrap with Bonferroni adjustment" described on page 402 of HM2015 and apply it to `stem_data`. Briefly discuss your findings.